DARPAN BANERJEE

Python Developer | Backend Developer | Flask | SQL | APIs | Automation
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Purulia, West Bengal, India

Professional Summary

Computer Science postgraduate with hands-on experience in Python development, backend web applications, SQL databases, APIs, and automation tools. Built real-world projects using Flask, SQLite, MySQL, and Python libraries. Currently working in technical training and system operations roles with practical exposure to ERP systems and software workflows. Strong problem-solving ability and eager to contribute as a Junior Python Developer.

Technical Skills

- **Programming Languages:** Python, SQL, C, JavaScript, HTML, CSS
- **Frameworks / Libraries:** Flask, Pandas, NumPy, Scikit-learn
- **Database:** MySQL, SQLite
- **Core Concepts:** OOP, File Handling, REST API, CRUD Operations
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook

Education

B.Sc. in Computer Science (SGPA: 8.87/10) *Sep 2018 – Jul 2021*
Manbhum Mahavidhlaya, Manbazar, Purulia, West Bengal

M.Sc. in Computer Science (CGPA: 8.88/10) *Sep 2021 – Aug 2023*
Sidho-Kanho-Birsha University, Purulia, West Bengal

TCS National Qualifier Test (NQT-2026) – 71.36%
Foundation Section: **70.97%** | Advance Section: **76.19%** | Programming Section: **67.70%**

Certifications

Diploma in IT (12 Months) | **Mathematics in Deep Learning** | **Data Analysis using Python**

PROFESSIONAL EXPERIENCE

Python Trainer & Application Developer

ABS Academy, Durgapur, WB
(*March 2025 – Present)*

- Delivered hands-on training in Python programming, SQL, and real-world projects
- Developed Student ERP system for record management and reporting
- Built Python-based tools for automation of academic workflows
- Guided students in backend development using Flask and databases
- Assisted in debugging, logic building, and project deployment basics

Technical Support & Network Administrator

Sarthak Educational Trust-16678 (TCS Examination Centre), Durgapur, WB
(March 2025 – Present)

- Managed network infrastructure and system configuration for large-scale online examinations
- Monitored system performance and ensured zero downtime during critical operations
- Performed troubleshooting of network connectivity, system security, and hardware issues
- Coordinated technical setup to maintain seamless examination workflow

Projects

Academic ERP Management System | *Python, Flask, SQLite, Pandas, HTML, CSS, GitHub*

- Designed and developed a full-stack Academic ERP application to streamline student record management, attendance tracking, and departmental operations.
- Built complete CRUD functionality for managing student profiles, academic details, and attendance records using SQLite database.
- Automated report generation and Excel export using Pandas to improve administrative reporting efficiency.
- Structured the application using modular Flask architecture (routes.py, auth.py, models.py) for maintainability and scalability.
- Applied real-world software development practices including Git version control and GitHub repository management.
- GitHub: https://github.com/Darpan2000-web/academic-erp-system

Automation Testing Project | *Python, Selenium, PyTest*

- Developed a test automation framework using Python and Selenium WebDriver
- Implemented Page Object Model (POM) for maintainable test structure
- Automated login functionality with positive and negative test cases
- Integrated PyTest for test execution and HTML reporting
- Implemented screenshot capture on test failure
- GitHub: https://github.com/Darpan2000-web/automation-testing-framework

Weather Data Pipeline & Visualization System | *Python, Flask, SQL, HTML, CSS, API*

- Transformed raw JSON data into structured format using Python and Pandas
- Stored processed data in a SQLite database for efficient querying and management
- Developed a Flask-based web application to dynamically display weather insights
- Built a modular and scalable pipeline architecture (Extract → Transform → Load)
- Utilized location-based data fetching to support dynamic user queries
- Strengthened understanding of data engineering concepts, backend development, and API handling
- GitHub: https://github.com/Darpan2000-web/weather-data-pipeline

Salary-Analysis-Career-Recommendation-System | *Python, Pandas, NumPy, Matplotlib*

- Analyzed 250K+ records to identify key factors influencing salary using EDA and statistical techniques.
- Executed ANOVA and normality testing to validate significant features affecting salary.
- Discovered that experience, job role, education, and location significantly impact salary, while industry does not develop a salary-based profiling system to identify required experience, skills, and roles for different salary
- Designed an ML-ready framework for salary prediction and career recommendation system
- GitHub: https://github.com/Darpan2000-web/Salary-Analysis-Career-Recommendation-System

House Price Prediction (Machine Learning) | *Python, Scikit-learn, Pandas, NumPy, Matplotlib*

- Completed data cleaning, feature engineering, and model evaluation
- Achieved strong predictive performance using cross-validation techniques through regression.
- Achieved R^2 Score of 0.67 using Linear Regression and 0.50 using optimized Random Forest model
- Implemented end-to-end ML pipeline from data preprocessing to model evaluation
- GitHub: https://github.com/Darpan2000-web/House_Price_Prediction_Project

Data Analysis Projects | *Power BI, Excel, MySQL*

- Enforced KPI metrics (230M sales, 286K profit) and region-wise analysis to support decision-making
- Applied data transformation and modeling techniques for accurate reporting
- Enabled dynamic filtering using slicers (city, state, sub-category) for deep-dive analysis
- Delivered actionable insights on regional performance and product category trends
- GitHub: https://github.com/Darpan2000-web/Sales-Data-Analysis-for-a-Retail-Store

Frontend Web Development

- Developed Amazon and Netflix UI clones using HTML, CSS, JavaScript
- GitHub: https://github.com/Darpan2000-web/webpage_clone

Additional Information

- Open to relocation across India for Data Analyst opportunities
- Available for full-time corporate roles
- Actively building advanced analytics portfolio (SQL, Python, Power BI)
- Strong analytical and problem-solving mindset with continuous upskilling focus